Intro Lab 2: Minesweeper Implementation

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Current Lab 2 result

This Lab 2 version implements the core interactions of a playable Minesweeper game. The current scope contains two rectangular difficulty modes - Simple and Advanced - with a complete interaction flow from starting a game to winning or hitting a mine.

How to use

Select a difficulty, left-click to reveal cells, and right-click to mark or unmark suspected mines. Use PAUSE to suspend the game and RESET to start again. The player wins by revealing all safe cells.

Interaction design and implemented functionality

• **Reveal**: left-click a cell to reveal it. Empty areas automatically expand to connected safe cells.

• **Marking**: right-click an unrevealed cell to place or remove a flag; the marked counter updates immediately.

• **Feedback**: adjacent mine counts use different colors, counters update during play, and dialogs communicate undo, loss and victory states.

• **Error recovery**: in Simple mode, the first mine hit can be undone once.

• **Game control**: timer, pause, reset and difficulty selection are available throughout play.

• **Difficulty**: Simple uses a 9 x 9 board with 15 mines; Advanced uses a 12 x 12 board with 30 mines.

Implementation overview

• **Model**: BoardModel stores mines, revealed cells and marked cells in two-dimensional boolean arrays.

• **Mine generation**: mines are generated after the first click and exclude the clicked cell, making the first reveal safe.

• **Automatic expansion**: an empty cell recursively reveals neighboring safe cells until numbered boundary cells are reached (a flood-fill-style reveal).

• **MVC structure**: Game coordinates interaction and game state; BoardUi presents visual feedback; MiniTimer uses a Swing Timer for elapsed time and pause/resume.

Scope and next iteration

Lab 2 focuses on stable Simple and Advanced rectangular modes. A later iteration will refine the UI and add the Innovative hexagonal-board mode while preserving the interaction logic developed here.

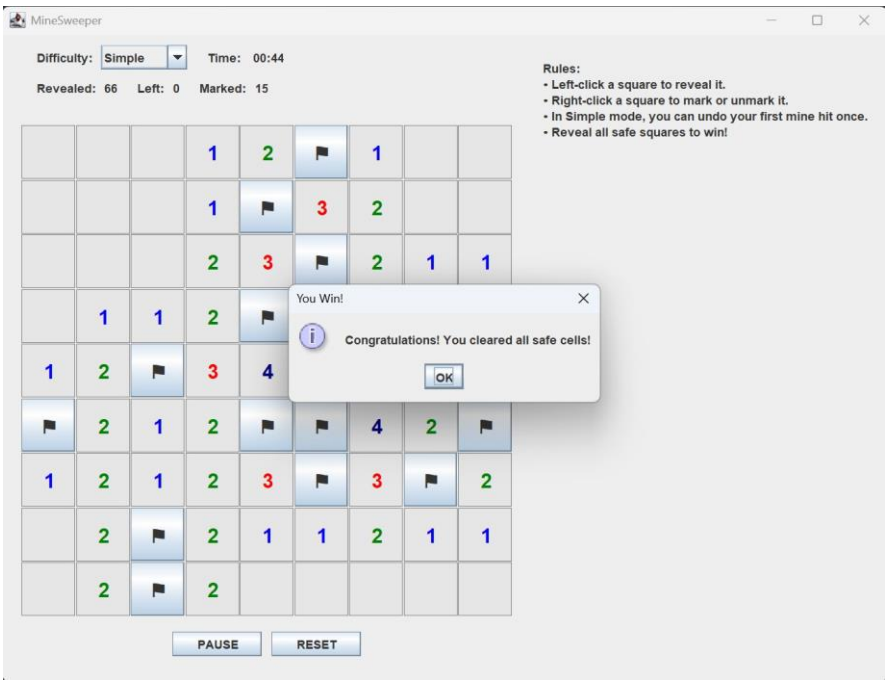
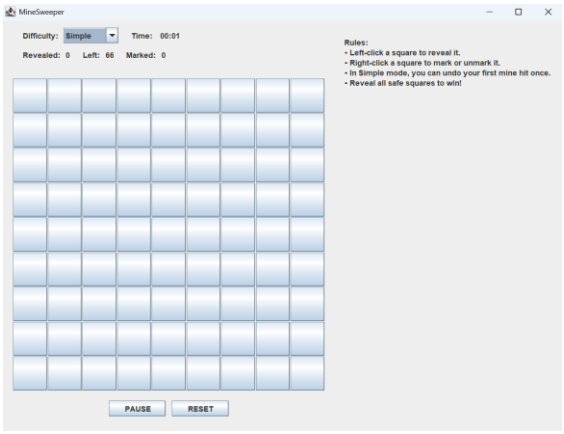


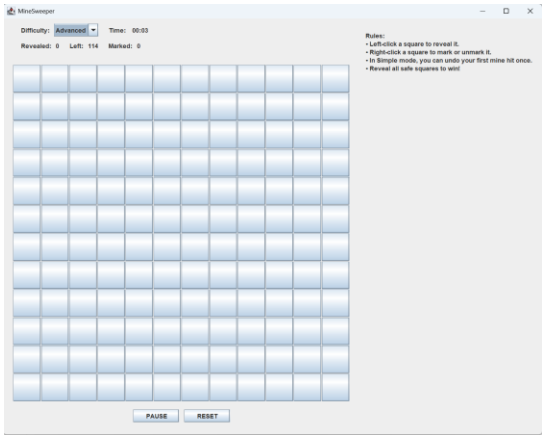
Figure 1. Completed Simple-mode game showing colored number feedback, flags, counters and the victory dialog.

Interaction states and difficulty modes

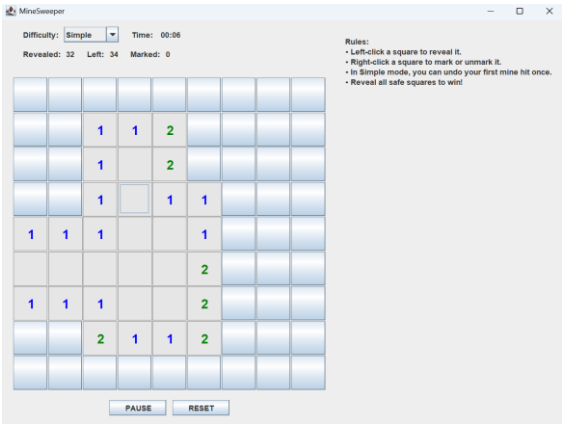
The screenshots below document the main interaction states tested in the Lab 2 implementation. They show how the interface responds to different user actions rather than only presenting the final visual layout.



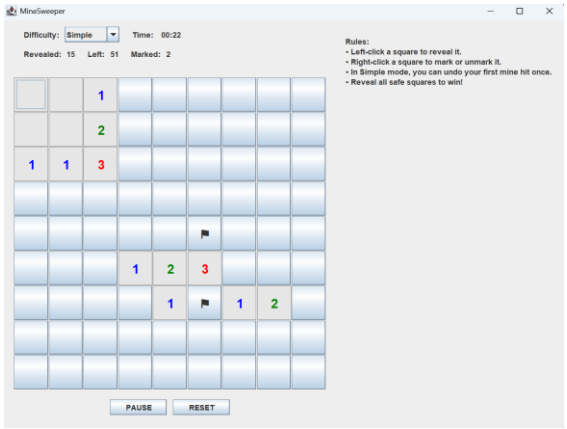
Simple mode: *9 x 9 board, 15 mines, 66 safe cells.*



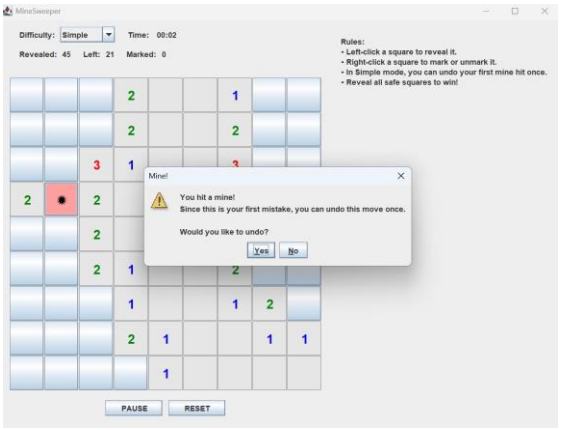
Advanced mode: *12 x 12 board, 30 mines, 114 safe cells.*



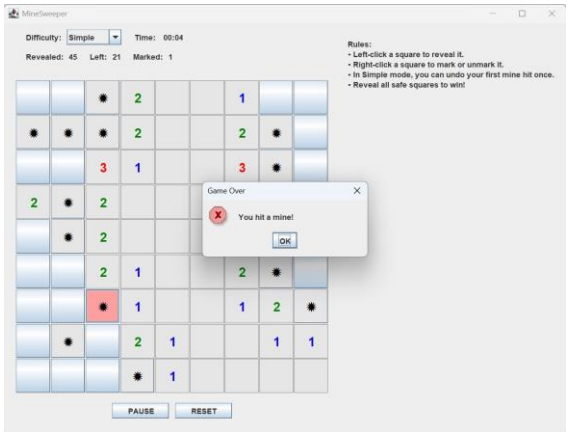
Reveal interaction: *safe cells and adjacent-mine numbers are displayed.*



Mark interaction: *right-clicked cells are marked with flags.*



Error recovery: *the first Simple-mode mine hit offers a one-time undo.*



Game-over feedback: *mines are revealed and the hit mine is highlighted.*

Testing and validation

I tested difficulty switching, safe first click, recursive empty-area expansion, repeated flag/unflag actions, protection of marked cells during automatic expansion, pause/resume, reset, the one-time Simple-mode undo, later mine-hit game over, counter updates, and the win condition when all safe cells are revealed.

Current limitations and next steps

This version prioritizes stable core interaction over final visual polish. The one-time mine-hit undo is intentionally limited to Simple mode. The current boards are rectangular; UI refinement and the Innovative hexagonal-board version are planned for a later iteration.